

## POLARIZED ABSORPTION, FLUORESCENCE AND PHOSPHORESCENCE SPECTRA OF CORONENE IN TRIPHENYLENE MATRIX AT 4.2°K

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Highly resolved absorption, fluorescence and phosphorescence spectra of coronene molecules dissolved in a triphenylene solid-state matrix have been observed at 4.2°K. It is found that each spectrum is composed of two sub-spectra with different polarization: one of them, subspectrum I, consists of the bands allowed via intramolecular vibronic interactions and the other, subspectrum II, contains not only a remarkably enhanced 0,0 transition band in contrast to those observed for *n*-paraffine solid solutions but also comprises some new bands. From the results of the analysis of these spectra it is suggested that the coronene molecules in a triphenylene matrix are distorted into a symmetry lower than the  $D_{6h}$  symmetry of free coronene molecules.

### 1. Introduction

As is well known, absorption, fluorescence and phosphorescence spectra of coronene in *n*-paraffine solid solutions have been studied as a typical example of the Shpol'skii effect [1-10]. The vibronic bands were analyzed in terms of symmetry-forbidden transitions [10, 11], which can be allowed via vibronic perturbations of some  $e_{2g}$  vibrations\*. In order to complete the analysis of the vibronic bands, some detailed studies of fundamental vibrations are needed; only several Raman lines have been reported [10, 12]. For this purpose the polarization measurements of the vibronic bands are useful [13-16].

In this paper, the polarized absorption, fluorescence and phosphorescence spectra of coronene molecules dissolved in triphenylene crystals are studied, and the vibronic bands of these spectra are analyzed.

### 2. Experimental

The coronene employed in this study was highly purified by the method previously reported [17]. Tri-

phenylene, as a matrix, was recrystallized and sublimed, and then thoroughly purified by the zone refining method. The mixed crystal sample was prepared by a melt grown technique as follows: the purified coronene was mixed with triphenylene on a quartz plate (11 mm × 17 mm × 1 mm) and it was covered with another quartz plate. In the presence of nitrogen gas, the sandwiched sample was melted and then rapidly cooled down from the 11 mm edge side of the quartz plate by a blow of cooled nitrogen gas. The mixed crystals were almost uniformly grown to a direction parallel to the 17 mm edge. This direction is considered to be parallel to the *c* axis since pure crystals of triphenylene are elongated in the *c* direction [18]. The sandwiched sample was mounted in a liquid helium cryostat; the 17 mm edges were always set vertical. The concentration of coronene in the solid solution was about  $10^{-4}$  mole/mole.

For the measurements of the emission spectra, an ultra high pressure mercury arc was employed as an excitation light source. Light emitted from the arc passed an Aminco grating monochromator and then the exciting light was focussed on a face of the sample from a horizontal direction. The light beam emitted from the face of the sample to another horizontal direction was collected by a lens and focussed on the

\* Representations in  $D_{6h}$  are used throughout this paper. \*

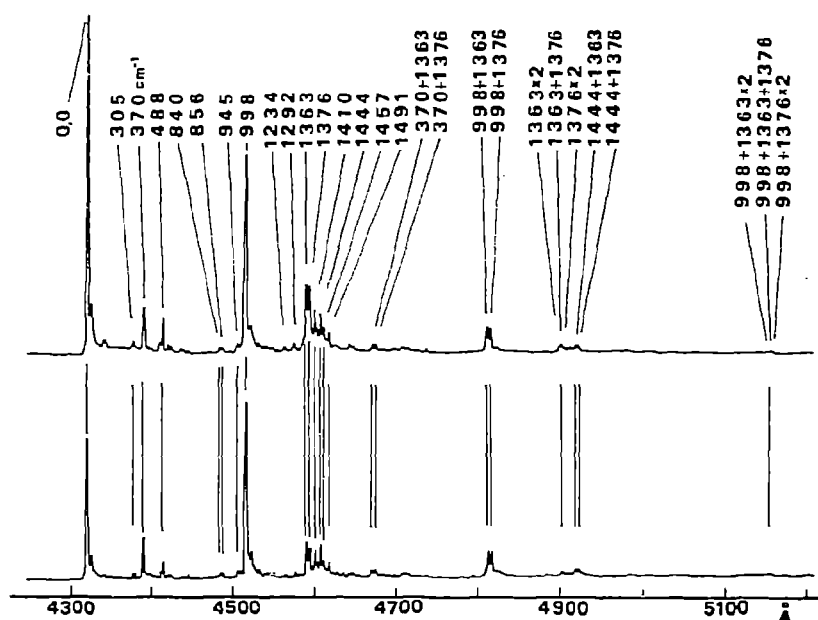


Fig. 1. The polarized fluorescence spectra of coronene in triphenylene at 4.2°K. The upper and lower spectra correspond to vertical and horizontal polarization, respectively. The highest energy band labelled 0,0 is at 4322.2 Å (23136  $\text{cm}^{-1}$ ).

entrance slit of a Spex 1700 II grating monochromator and then detected with a photomultiplier. For the polarization measurements a prism polarizer and a polaroid sheet were inserted in the exciting and the emitted light paths, respectively.

The  $S_0-S_1$  absorption spectra of coronene were recorded by the fluorescence or the phosphorescence excitation method. The optical arrangement was very similar to that for the measurements of the emission spectra: only the monochromators were exchanged in order to obtain the desired resolving power, except for the natural replacement of the mercury arc by a xenon arc.

A low pressure mercury arc was used for wavelength calibration. It was estimated that the wavelength could be determined to  $\pm 0.2$  Å.

### 3. Results and discussion

#### 3.1. The fluorescence spectra

The polarized fluorescence spectra of coronene in

triphenylene at 4.2°K are shown in fig. 1 with the frequencies of several of the more prominent vibronic bands labelled. The highest energy band at  $23136 \text{ cm}^{-1}$  coincides with the lowest energy band in the absorption spectrum, as illustrated in fig. 2. Therefore, it is assigned as the 0,0 transition band. The polarization intensity ratios, vertical:horizontal, are independent of the wavelength, 313 nm or 365 nm, and also of the polarization of the exciting light. The difference of the polarization intensity ratios for the more prominent bands enables us to divide the spectra into two subspectra, table 1: subspectrum I<sub>f</sub> with the values of about 1.1 and subspectrum II<sub>f</sub> with those of about 2.

The subspectrum I<sub>f</sub> is composed of many prominent bands, such as, 370, 998, 1444, 2358 and  $2378 \text{ cm}^{-1}$ , which corresponds to the very intense bands observed for *n*-octane solid solution, 374 ( $e_{2g}$ ), 1001 ( $e_{2g}$ ), 1444 ( $e_{2g}$ ), 1001 + 1352 ( $a_{1g}$ ) and 1001 + 1370 ( $a_{1g}$ )  $\text{cm}^{-1}$  [10]. The assumed two totally symmetric vibrations of coronene for the *n*-octane solution, 1352 and  $1370 \text{ cm}^{-1}$ , really correspond to the very intense Raman lines, 1354 and  $1369 \text{ cm}^{-1}$  [10]. An analogous analysis of the 2358 and  $2378 \text{ cm}^{-1}$

Table 1

The subspectrum  $I_f$  and the subspectrum  $II_f$  of the polarized fluorescence spectra of coronene in triphenylene.  $\Delta\nu = \nu_{0,0} (23136 \text{ cm}^{-1}) - \nu$ .  $I_v/I_h$  is the polarization intensity ratio, vertical:horizontal

| Subspectrum $I_f$ |           |                         |             |           |                       | Subspectrum $II_f$ |           |                                         |
|-------------------|-----------|-------------------------|-------------|-----------|-----------------------|--------------------|-----------|-----------------------------------------|
| $\Delta\nu$       | $I_v/I_h$ | assignment              | $\Delta\nu$ | $I_v/I_h$ | assignment            | $\Delta\nu$        | $I_v/I_h$ | assignment                              |
| 370               | 1.1       | 370 ( $e_{2g}$ )        | 2358        | 1.0       | 998 + 1363            | 0                  | 2.4       | 0,0 ${}^1B_{2u} \rightarrow {}^1A_{1g}$ |
| 856               | 1.2       | 856                     | 2378        | 1.1       | 998 + 1376            | 305                | 1.6       | 305                                     |
| 945               | 1.2       | 945                     | 2595        | 1.0       | 1234 + 1363           | 488                | 1.9       | 488 ( $a_{1g}$ )                        |
| 998               | 1.1       | 998 ( $e_{2g}$ )        | 2612        | 1.0       | 1234 + 1376           | 1363               | 1.9       | 1363 ( $a_{1g}$ )                       |
| 1234              | 1.4       | 1234 ( $e_{2g}$ )       | 2654        | 1.0       | 1292 + 1363           | 1376               | 1.8       | 1376 ( $a_{1g}$ )                       |
| 1292              | 1.1       | 1292 ( $e_{2g}$ )       | 2671        | 1.0       | 1292 + 1376           | 2724               | 1.6       | 1363 $\times$ 2                         |
| 1410              | 1.4       | 1410 ( $e_{2g}$ )       | 2768        | 1.0       | 1410 + 1363           | 2739               | 1.6       | 1363 + 1376                             |
| 1444              | 1.1       | 1444 ( $e_{2g}$ )       | 2782        | 1.0       | 1410 + 1376           | 2752               | 1.6       | 1376 $\times$ 2                         |
| 1457              | 1.3       | 1457                    | 2808        | 1.0       | 1444 + 1363           |                    |           |                                         |
| 1491              | 1.2       | 1491                    | 2822        | 1.0       | 1444 + 1376           |                    |           |                                         |
| 1627              | 1.1       | 1627 ( $e_{2g}$ )       | 3720        | 0.9       | 998 + 1363 $\times$ 2 |                    |           |                                         |
| 1733              | 1.0       | 370 + 1363 ( $a_{1g}$ ) | 3736        | 0.9       | 998 + 1363 + 1376     |                    |           |                                         |
| 1750              | 0.9       | 370 + 1376 ( $a_{1g}$ ) | 3754        | 0.9       | 998 + 1376 $\times$ 2 |                    |           |                                         |

bands yields two kinds of totally symmetric vibrations, 1360 and 1380  $\text{cm}^{-1}$ , for the coronene molecules in the triphenylene matrix. Most of the bands in the subspectrum  $I_f$  were assigned as some  $e_{2g}$  fundamentals +  $n \times 1363 (a_{1g}) + m \times 1376 (a_{1g})$  in terms of the selection rule for the symmetry forbidden  ${}^1B_{2u} \rightarrow {}^1A_{1g}$  transition, where  $n$  and  $m$  are 0, 1 or 2. Table 1 shows the results.

The subspectrum  $II_f$  contains the 0,0 band, which has the largest polarization intensity ratio, 2.4, and also comprises a few bands, 488, 1363, and 1376  $\text{cm}^{-1}$ , which accord well with the three strongest Raman lines, 487, 1354, and 1369  $\text{cm}^{-1}$ , possibly the most Raman active  $a_{1g}$  vibration [10]. These three bands are identically polarized with the 0,0 band so that they have a totally symmetric character, and therefore they are assigned as  $a_{1g}$  vibrations. Among them the larger two  $a_{1g}$  vibrations, 1363 and 1376  $\text{cm}^{-1}$ , agree well with the postulated  $a_{1g}$  vibrations in the analysis of the subspectrum  $I_f$ . Moreover, overtone bands and combination bands of these two  $a_{1g}$  vibrations are found in the subspectrum  $II_f$  at 2724, 2739, and 2752  $\text{cm}^{-1}$  with fairly large polarization intensity ratios of about 1.6. The appearance of the  $a_{1g}$  bands reveals that the selection rule for the symmetry forbidden  ${}^1B_{2u} \rightarrow {}^1A_{1g}$  transition has broken down.

### 3.2. The absorption spectra

The absorption spectra of coronene in the tri-

phenylene matrix at 4.2°K were obtained as the excitation spectra of the fluorescence or the phosphorescence. An unpolarized but well resolved spectrum, fig. 2, was obtained by the observation of the intense fluorescence band at 22138  $\text{cm}^{-1}$  (998  $\text{cm}^{-1}$  band in fig. 1). The observation wavelength did not affect the intensity distribution of the vibronic bands even if the phosphorescence was excited. The polarization intensity ratios of the vibronic bands were also measured and are listed in table 2. The subspectra  $I_a$  and  $II_a$  in the table are the analogues of the subspectra  $I_f$  and  $II_f$  of the fluorescence, respectively; the subspectrum  $I_a$  is subject to the forbidden selection rule and the subspectrum  $II_a$  to the allowed one. However, only one additional  $a_{1g}$  vibration, 1381  $\text{cm}^{-1}$ , is found.

### 3.3. The phosphorescence spectra

The polarized phosphorescence spectra of coronene in triphenylene at 4.2°K are illustrated in fig. 3. The highest energy band is located at 19142  $\text{cm}^{-1}$ . The vibronic bands are also divided into two subspectra,  $I_p$  and  $II_p$ , table 3. Subspectrum  $I_p$  consists of the bands with polarization intensity ratios of about 1.1. The more prominent bands, 371, 484, 1156, 1230, 1409, 1444 and 1636  $\text{cm}^{-1}$ , agree well with those observed for the *n*-octane solution, 367, 484, 1154, 1224, 1407, 1439 and 1633  $\text{cm}^{-1}$  [10].

Subspectrum  $II_p$  includes the highest energy band

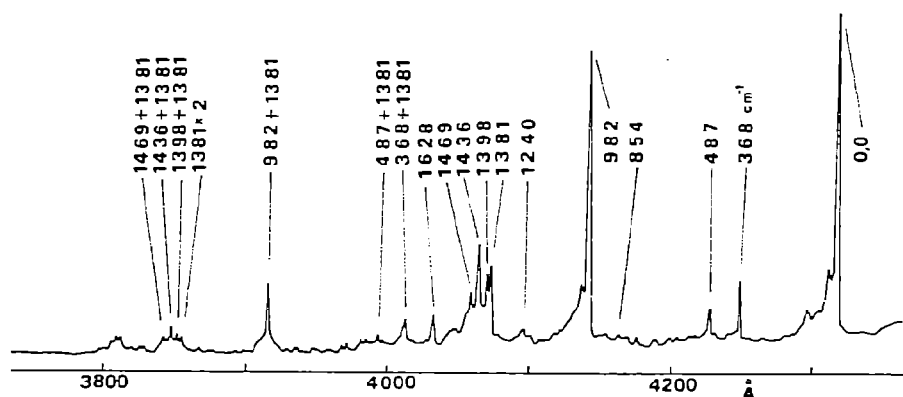


Fig. 2. The unpolarized first singlet-singlet absorption spectrum of coronene in triphenylene at 4.2°K by fluorescence excitation. The lowest energy band labelled 0,0 is at 4322.2 Å (23136 cm<sup>-1</sup>).

Table 2

The subspectrum I<sub>a</sub> and the subspectrum II<sub>a</sub> of the polarized absorption spectra of coronene in triphenylene.  $\Delta\nu = \nu - \nu_{0,0}$  (23136 cm<sup>-1</sup>).  $I_v/I_h$  is the polarization intensity ratio, vertical:horizontal

| Subspectrum I <sub>a</sub> |           |                               | Subspectrum II <sub>a</sub> |           |                                                                 |
|----------------------------|-----------|-------------------------------|-----------------------------|-----------|-----------------------------------------------------------------|
| $\Delta\nu$                | $I_v/I_h$ | assignment                    | $\Delta\nu$                 | $I_v/I_h$ | assignment                                                      |
| 368                        | 1.3       | 368 (e <sub>2g</sub> )        | 0                           | 2.4       | 0,0 <sup>1</sup> A <sub>1g</sub> → <sup>1</sup> B <sub>2u</sub> |
| 982                        | 1.2       | 982 (e <sub>2g</sub> )        | 487                         | 2.0       | 487 (a <sub>1g</sub> )                                          |
| 1398                       | 1.1       | 1398 (e <sub>2g</sub> )       | 1381                        | 2.4       | 1381 (a <sub>1g</sub> )                                         |
| 1436                       | 1.2       | 1436 (e <sub>2g</sub> )       | 2759                        | —         | 1381 × 2                                                        |
| 1469                       | 1.1       | 1469 (e <sub>2g</sub> )       |                             |           |                                                                 |
| 1628                       | 1.0       | 1628 (e <sub>2g</sub> )       |                             |           |                                                                 |
| 1749                       | 1.0       | 368 + 1381 (a <sub>1g</sub> ) |                             |           |                                                                 |
| 2361                       | 1.2       | 982 + 1381                    |                             |           |                                                                 |
| 2780                       | —         | 1398 + 1381                   |                             |           |                                                                 |
| 2814                       | —         | 1436 + 1381                   |                             |           |                                                                 |
| 2844                       | —         | 1469 + 1381                   |                             |           |                                                                 |

assigned as the 0,0 band and it comprises in addition some new bands not observed for the *n*-paraffine solid solutions [1, 4, 10], such as 1363 and 1376 cm<sup>-1</sup>, which are just coincident with the a<sub>1g</sub> vibrations found in the subspectrum II<sub>f</sub> and also found in the subspectrum I<sub>f</sub>.

### 3.4. General remarks and conclusion

The observed full widths at half maxima for the prominent bands are only 7 cm<sup>-1</sup>. Such sharpness of the spectral response lines indicates that the coronene molecules may be satisfactorily inserted in the tri-

phenylene crystals, which is very probable since the molecular radii of both planar hydrocarbons are almost equal. Although the host crystals have four molecules in each unit cell [18], the four sites are equivalent under the crystal symmetry operations and therefore all the coronene molecules have the same energy levels. Actually, the 0,0 band of the fluorescence spectra coincides with that of the absorption spectra and no Shpol'skii multiplets are observed, figs. 1–3. However, the four sites are spatially inequivalent for the optical measurements, and thus they provide four kinds of molecular orientation for the guest molecules. Fortunately, the crystal *c* axis

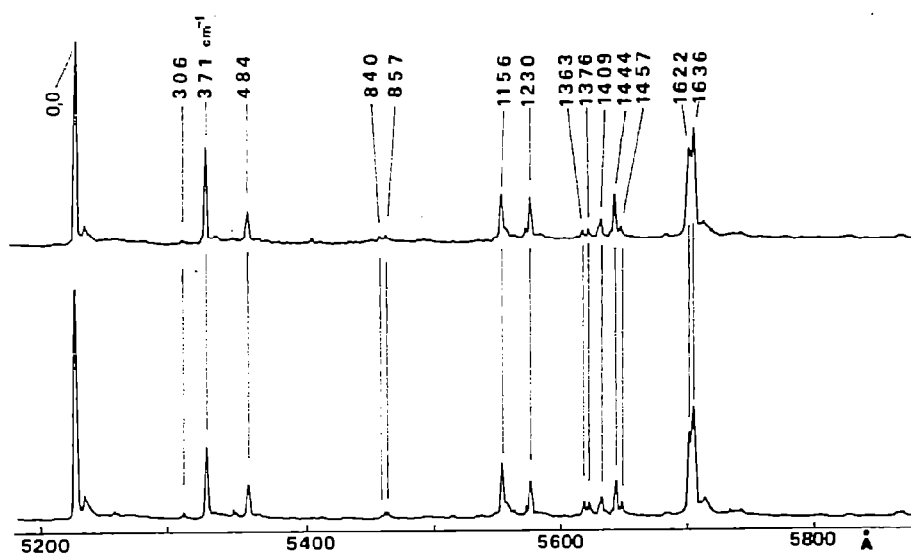


Fig. 3. The polarized phosphorescence spectra of coronene in triphenylene at 4.2°K. The upper and lower spectra correspond to vertical and horizontal polarization, respectively. The highest energy band labelled 0,0 is at 5224.1 Å (19142 cm<sup>-1</sup>).

Table 3

The subspectrum I<sub>p</sub> and the subspectrum II<sub>p</sub> of the polarized phosphorescence spectra of coronene in triphenylene.  $\Delta\nu = \nu_{0,0}$  (19142 cm<sup>-1</sup>) -  $\nu$ .  $I_v/I_h$  is the polarization intensity ratio, vertical:horizontal

| Subspectrum I <sub>p</sub> |           |                         | Subspectrum II <sub>p</sub> |           |                                                                 |
|----------------------------|-----------|-------------------------|-----------------------------|-----------|-----------------------------------------------------------------|
| $\Delta\nu$                | $I_v/I_h$ | assignment              | $\Delta\nu$                 | $I_v/I_h$ | assignment                                                      |
| 371                        | 1.1       | 371 (e <sub>2g</sub> )  | 0                           | 0.8       | 0,0 <sup>3</sup> B <sub>1u</sub> → <sup>1</sup> A <sub>1g</sub> |
| 484                        | 1.0       | 484                     | 306                         | 0.7       | 306                                                             |
| 1156                       | 1.0       | 1156                    | 1363                        | 0.8       | 1363 (a <sub>1g</sub> )                                         |
| 1230                       | 1.2       | 1230 (e <sub>2g</sub> ) | 1376                        | 0.8       | 1376 (a <sub>1g</sub> )                                         |
| 1409                       | 1.1       | 1409 (e <sub>2g</sub> ) |                             |           |                                                                 |
| 1444                       | 1.2       | 1444 (e <sub>2g</sub> ) |                             |           |                                                                 |
| 1457                       | 1.0       | 1457                    |                             |           |                                                                 |
| 1622                       | 1.1       | 1622 (e <sub>2g</sub> ) |                             |           |                                                                 |
| 1636                       | 1.1       | 1636 (e <sub>2g</sub> ) |                             |           |                                                                 |

has a common angle, 39°, to the planes of molecules in the different sites [18]. Therefore, the angle of the transition dipole vector for each vibronic band to the *c* axis can be related to the observed polarization intensity ratio,  $I_v/I_h$  in tables 1–3. The ratios for the subspectra I<sub>f</sub>, I<sub>a</sub>, and I<sub>p</sub> have a common value of about 1.1. Therefore, it seems that the vibronic perturbations by e<sub>2g</sub> vibrations are responsible for these

subspectra I. While, the ratios for the subspectra II<sub>f</sub> and II<sub>a</sub> have a high value of about 2, which may be considered as resulting from crystal field mixing of the lowest excited singlet state with <sup>1</sup>E<sub>1u</sub> states and also possibly with <sup>1</sup>A<sub>2u</sub> states. On the other hand, in the phosphorescence spectra the value for the subspectrum II<sub>p</sub> is about 0.8, which is different from the other values, for the singlet–singlet transition bands, 1,1 and 2. It can be explained in terms of the mechanisms that make the forbidden transitions allowed; the 0,0 phosphorescence of the aromatic hydrocarbon may be observed as a result of some out-of-plane spin–orbit interactions, whereas the 0,0 fluorescence must be enhanced by intensity borrowing from some allowed singlet–singlet transitions.

The difference of the polarization intensity ratios enabled us to assign the observed vibronic bands, however, the intensity distribution of the vibronic bands also gave us some important information. Although the lowest singlet–singlet and the lowest singlet–triplet transitions of a free coronene molecule are symmetry forbidden, the 0,0 transition intensities in figs. 1–3 are remarkably enhanced in the vibronic bands. Analogous phenomena have been reported for the S<sub>0</sub>–S<sub>1</sub> transition: in a recrystallized *n*-heptane

solution the intensity ratio  $I_{0,0}/I_{370}$  amounts to a value of 2.05 [8] and in an *n*-heptane solution under a high pressure of 30 kbar the intensity ratio  $I_{0,0}/I_{1000}$  is increased to about 0.5 [9], in contrast to the values obtained under usual conditions,  $I_{0,0}/I_{370} = 0.4-0.6$  and  $I_{0,0}/I_{1000} = 0.05-0.1$ .

In our measurements for *n*-octane solutions at 77°K,  $I_{0,0}/I_{370}$  is 0.15 and  $I_{0,0}/I_{1000}$  is 0.03. Whereas for the present mixed crystal fluorescence,  $I_{0,0}/I_{370}$  amounts to the largest value of 7.0 and also  $I_{0,0}/I_{1000}$  to 1.7. Such enhancement of 0,0 transition bands is also observed in the phosphorescence spectra as can be seen from fig. 3 by comparison with those for *n*-paraffine solutions published in the literature [2, 10].

It is therefore shown that the symmetry forbidden electronic transitions of coronene molecules become allowed to a considerable extent in the triphenylene matrix. The breakdown of the symmetry forbidden selection rule is conclusive, since some totally symmetric vibration bands, for example 1363 and 1376  $\text{cm}^{-1}$ , were found both in fluorescence and in phosphorescence. This conclusion seems to be curious and difficult to understand because the triphenylene molecules are neutral and expected to be non-polar. However, the symmetry lowering of coronene molecules in crystals of themselves has been shown theoretically [19]. An analogous effect of the host triphenylene molecules on dissolved coronene molecules is also possible; the mobile  $\pi$ -electrons are polarized so as to minimize the total energy of the system and

then the highly symmetric coronene molecules lose their  $D_{6h}$  symmetry into a lower symmetry, most probably  $D_{2h}$ , where the symmetry forbidden transitions in  $D_{6h}$ ,  $B_{2u}-A_{1g}$  and  $B_{1u}-A_{1g}$ , become allowed.

## References

- [1] E.J. Bowen and B. Brocklehurst, J. Chem. Soc. (1954) 3875.
- [2] E.J. Bowen and B. Brocklehurst, J. Chem. Soc. (1955) 4320.
- [3] E.V. Shpolskii and L.A. Klimova, Soviet Phys. Doklady 1 (1956) 782.
- [4] E.V. Shpolskii, Soviet Phys. Uspekhi 3 (1960) 372.
- [5] E.V. Shpolskii, Soviet Phys. Uspekhi 6 (1963) 411.
- [6] G.M. Svishchev, Opt. Spectry. 16 (1964) 200.
- [7] G.M. Svishchev, Opt. Spectry. 18 (1965) 350.
- [8] E.G. Moysya, Opt. Spectry. 21 (1966) 59.
- [9] D.E. Hein and H.W. Offen, Mol. Cryst. 5 (1969) 217.
- [10] K. Ohno, T. Kajiware and H. Inokuchi, Bull. Chem. Soc. Japan 45 (1972) 996.
- [11] J.W. Sidman, J. Chem. Phys. 23 (1955) 1365.
- [12] N. Mentzel and H. Zimmermann, Ber. Bunsenges. Physik. Chem. 67 (1963) 582.
- [13] D.S. McClure, J. Chem. Phys. 22 (1954) 1012.
- [14] T.N. Bolotnikova, Opt. i Spektroskopiya 7 (1959) 44.
- [15] T. Pavlopoulos and M.A. El-Sayed, J. Chem. Phys. 41 (1964) 1082.
- [16] M.A. El-Sayed, J. Chem. Phys. 43 (1965) 2864.
- [17] T. Kajiware, K. Ohno, S. Iwashima and H. Inokuchi, Bull. Chem. Soc. Japan 42 (1969) 2734.
- [18] A. Klug, Acta Cryst. 3 (1950) 165.
- [19] K. Ohno and H. Inokuchi, Theoret. Chim. Acta 26 (1972) 331.